

MLO OBSERVATORY GUIDE

UNOFFICIAL



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DISCLAIMER

This "guide" is **NOT** intended to be an observing "manual", and is not endorsed in any way by the Observatory Director. Before using these notes, a thorough familiarity with both the 'Mount Laguna Observatory Handbook' and the 'Instructions for Safety Training' is expected, as well as a full check-out by a "certified" SDSU faculty member. After that, these notes may provide some useful reminders and suggestions.

This document was prepared by Chuck Horst for his personal use as a guide to operating the telescopes at Mount Laguna Observatory. If you have been provided a copy by Chuck or anyone else you are welcome to use it with the understanding that no assurance of accuracy is implied.

1) LOGISTICS

- a) Altitude- 1859 meters
- b) Latitude- 32.84 (32:50.4)
- c) Longitude West- 116.426 (116:25.6)(7.76173 decimal hours)

2) PHONE NUMBERS

- a) Dorm (619) 470-3005
- b) 40" (619) 473-3001
- c) Andrew Farrow (858)353-4275

40" GUIDE



40" GUIDE

1) LOGISTICS

- a) PRINTER- 172.23.50.10

2) TELESCOPE SPECIFICATIONS

- a) F-RATIO
 - i) f/7.6 prime; f/13.5 cassegrain; f/33 coude

3) ENTER DOME

- a) Lock top bolt of door behind you
- b) Turn on (4) switches to immediate right of door after entering
 - i) These switches provide power to the telescope and lights.

4) OBSERVATORY FLOOR

- a) Walk up to observatory floor
 - i) Light switch inside 1st door if needed
- b) Observing floor
 - i) Lights should be on, if not switch is on right hand wall
 - ii) Red lights are controlled by rheostat
- c) At telescope
 - i) Check that instrument of choice is mounted
 - ii) For CCD2005 plug in guider power supply
 - iii) Turn on mirror cooling fans (summer)
 - (1) Cooling fans are probably not needed in the winter
 - (2) If fans are turned off or on expect about a 20 minutes for stabilization.
 - (3) Attach small cables at aft end of telescope
 - (4) Should hear a slight whirling sound
 - iv) Check LN2 if using CCD2005
 - (1) Check dewar log on West (far) wall
 - (2) Dewar should have been filled within last 52 hours to stay cold
 - (3) If dewar has not been filled within last 24 hours fill dewar to top off
 - (a) Stinger insert is in white cabinet on East wall center cabinet
 - (b) Insert stinger at aft end of gold dewar. Screw in insert at bottom of dewar (**CAUTION!** hand tight only)
 - (c) Place plastic tube on insert shaft only 3/8" up the shaft

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- (d) Make sure plywood is under telescope and the overflow exhaust is pointed over the plywood.
- (e) Open up valve fully on top of LN2 dewar
- (f) **VERY IMPORTANT** (DO NOT LEAVE THE TELESCOPE WHILE FILLING)
- (g) When full, LN2 will begin to flow out of overturn, turn off valve immediately.
- (h) Wait a few minutes for tube to warm up
- (i) Use blue gloves to remove insert and return gloves to cabinet when finished
- (j) **VERY IMPORTANT** (MOVE DEWAR OUT OF THE WAY) to not interfere with telescope in operation
- (k) Fill in log on clipboard on west wall
- (l) If LN2 tank is empty, bring a full one up from ground floor
- (4) If dewar is empty and the camera has warmed to room temperature fill the dewar as above and it will take approximately 3 hours for camera to come to temperature and stabilize.
- v) Check that the observing floor is clear, nothing to interfere with the movement of the telescope especially the ladder along the west wall.
- vi) Check the cabling is clear and properly placed on the guides
- vii) The hand pad should be out of the way and hung on the telescope pier

5) DATA ROOM

- a) Review observing log book
 - i) Look for recent problems/issues and resolutions
- b) Set up personal pc
 - i) Aladin
 - (a) Open Aladin from application
 - (b) Load finder list
 - (c) File/Open local file/browse for finder file
 - (d) double click list in right pane
 - (e) Load overlay image
 - (f) click "DSS" on top left
 - ii) JSkyCalc

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- (a) Open JSkyCalc from application
 - (b) Load object list
 - (c) Object list/Load New Object List/browse for object list
- iii) Browser
- (a) Seeing monitor at Palomar
<http://www.palomar.caltech.edu:8000/maintenance/index.tcl>
 - (b) Clear Sky Chart
<http://cleardarksky.com/c/MtLagOBCAkey.html?1>
 - (c) Accuweather <http://www.accuweather.com/us/ca/mount-laguna/91948/city-weather-forecast.asp?partner=netweather&traveler=0&u=1>
 - (d) Satellite
<http://observatories.hodar.com/palomar/index.html>
 - (e) Print out log sheets

CCD2005 GUIDE

1) SPECIFICATIONS

- a) Name-CCD2005
- b) Field of view- $\sim 13.5' \times 13.5'$
- c) Pixel scale CCD2005- $\sim 0.406''/\text{pixel}$
- d) Pixel scale guider is approx. $0.3''/\text{pixel}$

2) STARTUP STATIONS

- a) Reboot sequence
 - i) I have found it necessary to reboot both computers at the beginning of the run to assure the guide camera will work.
 - ii) Reboot ACE first the OWL
 - iii) Same password both OWL and ACE
- b) Left Monitor- **|ACE|**
 - i) Controls telescope moves, shutters, filters and dome controls
 - ii) Is to be always left up
 - iii) If ACE is not running
 - (1) Double click "ACE*release-date*" icon on desktop
 - (2) [Username]:Observer (Note: capital "O")
 - (3) [Password]:
 - (a) Note use "1" instead of "i"
 - (4) (LOGIN)
 - (5) If computer needs rebooted login is:
 - (a) [Username]: Observer; [Password]:
 - iv) Open windows on desktop and place on desktop
 - (1) <TELESCOPE> -> (Focus)
 - (2) <INSTRUMENTS> -> (Filter wheel)
 - (3) <TELESCOPE> -> (Center CCD)
 - (4) <OBSERVATORY> -> (Weather station)
 - v) Move guide camera mirror out of the way
 - (1) Initialize guide camera
 - (a) {Aux. Inst} <XY Stage> (Initialize)
 - (b) This is to make sure the visual on-screen location is synchronized with the actual mirror position
 - (2) Move mirror to center top just below top edge
 - (a) On-step method to move the mirror
 - (i) [Position]: 45 1224 (Goto)

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- (b) Step-by-step method to move the mirror
 - (i) Set move increment to 500 in yellow NESW box
 - (ii) Move pickoff mirror and watch red target
 - 1. N,N,N... E,E,E...
 - 2. Continue until centered and just below top edge
 - (iii) Reset move increment to 50
- c) Right Monitor- **[OWL]**
 - i) Used for exposure control
 - ii) Windows login: Observer Password:
 - iii) Start up OWL data taking software
 - (1) Double click "OWLversion" icon, this will open 2 windows {OWL} & {OWL LOG}
 - iv) Turn on CCD {OWLversion}
 - (1) <Controller> (Setup) (Apply)[Reset Controller →check](close)
 - (2) Set dewar temperature
 - (a) Best to do this early to permit the camera to achieve proper temperature and dwell for 20 minutes before starting science.
 - (b) <Supported Configuration> "SiDiode temperature" [Set array temp]: -125 (Run) (Close)
 - (c) Check that monitor reflects setting.
 - (i) SiDiode temperature [-125.0C]
 - (d) Record starting dewar temp in observing room log book
 - (e) To monitor temperature one time
 - (i) <Quick Actions> "Thermometer"
 - 1. Temperature will read out in {Controller} section
 - (f) To monitor temperature via plot
 - (i) <Quick Actions> "Thermometer with green scrawl" (RUN)
 - (3) Establish readout pattern
 - (a) <Supported Configuration> "Parallel Readout" (UR) (RUN) (Close)
 - (b) Check that monitor reflects setting
 - (i) Parallel readout [UR]
 - (4) If problems with above
 - (1) <Controller>(Setup)(Apply)[Reset controller→check](close)
 - (2) Then repeat setup of dewar temp and readout pattern
 - v) Open terminals

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- (1) <START><Programs><Cygwin-X><XWinServer>
- (2) <START> <Programs> <Cygwin-X> <Xterm>
- (3) Set data directory {XTERM}
 - (a) \$ cd /e/images/dleonard
 - (b) \$ mkdir 20110608 (date directory; use UTC convention- morning date)
- vi) Set up file Directory {OWLversion}<Image File Options>
 - (1) [Dir]: E:\images\dleonard\20110608
 - (a) Note capital "E" and backslash
 - (2) [File]: a1.fit
 - (a) [Save→check]
 - (b) [Increment Filename→check]
- vii) Set up auto display if desired
 - (1) CAUTION- having OWL load image in DS9 prevents user ability to change the scaling factors. Some may desire to load the images individually from IRAF
 - (2) {OWL}<Image analysis>[Display→check]
- viii) Header Info {OWLversion}<Image File Options>
 - (1) [Observer]: Horst
 - (2) Close header window
- ix) Start IRAF & DS9
 - (1) {XTERM} \$ xgterm -sb &
 - (2) {XGTERM} \$ cd /home/observer/iraf
 - (3) {XGTERM} \$ ds9 &
 - (4) Set zoom preference on ds9
 - (a) {ds9}<Edit>(Preferences)
 - (b) {Preferences} <Zoom> <Menu> (Zoom ¼)
 - (5) {XGTERM} \$ cl
 - (6) {XGTERM} ecl> cd /e/images/dleonard/20110608
 - (7) Check directory
 - (a) ecl> ls (note proceeding with a "!" bang runs unix commands. This shouldn't be necessary if IRAF was started correctly)
 - (b) Should be empty unless continuing run

3) SET UP OBSERVING LOGS

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- a) Be sure to include local sidereal time, FWHM, and peak values of selected target objects. In comments include when appropriate initial focus position, focus position with each focus adjustment, offsets, periodic weather assessment

4) CLOSE DOORS & TURN OFF LIGHTS

- a) Lights should be out in dome, stairs and hall outside of data room
- b) Doors should be closed at base of stairs, into data room hall, data room door and outside to catwalk

5) TEST CCD & TAKE BIAS

- a) Hopefully at this point the camera has reached -125 and dwelled for 20 minutes
- b) Clear chip
 - i) Check size of image should be $\sim 2196 \times 2048$ and
- c) Take bias
 - i) **|ACE|**
 - (1) Set name to be bias to automatically load into header object field
 - (a) {ACE} <RA> [Name]: bias
 - ii) **|OWL|**
 - (1) {OWL}<Exposure Options>
 - (a) [Open Shutter→unchecked]
 - (b) [Multiple Exposures→unchecked]
 - (2) [Exp. Time]: 0
 - (3) (EXPOSE)
- d) Check counts on ds9
 - i) Move cursor over field and look at [Value] numbers
 - ii) Should be $\sim 700-800$
- e) Check image size
 - i) Ecl>imhead a1 lo-
 - (1) Check that image size is [2200,2048]
- f) Take multiple bias (if pre-sunset take a lot ~ 20) **|OWL|{OWL|}**
 - i) <Exposure Options>
 - (1) [Multiple Exposures→check]
 - (2) [Multiple Exposures]: 19 or some number
 - ii) (EXPOSE)
 - iii) Repeat as necessary

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iv) [Multiple Exposures→uncheck]

6) OPEN SHUTTERS & MIRROR |ACE|

- a) Open shutters
 - i) <DOME> <open shutter> [Drop out→check][Shutter→check] (open)
 - ii) Wait for shutter and drop out to open before proceeding
 - iii) DOME shutter & drop out status light will say 'OPEN' on left
- b) Open Mirror Cover
 - i) Wait for shutter to fully open before opening the mirror cover. All precautions are to be made to assure nothing falls and damages the mirror.
 - ii) <TELESCOPE> <Open Mirror Cover> (yes)
- c) Check status lights
 - (1) All status lights on left hand side should be green
 - (2) DOME Status (Auto), RA Track (ON) and Dec. Track (OFF)

7) ENGAGE TRACKING & AUTO DOME |ACE|

- a) Engage tracking
 - i) <TELESCOPE> <track enable> (YES)
- b) Engage autodome
 - i) <DOME> <autodome> (YES)
 - ii) Go onto observing floor and check the telescope is pointed at the dome opening. Check both sides of the telescope by looking along the side of telescope out the dome

8) TAKE TWILIGHT FLATS

- a) Start setting up 30min before sunset
- b) Take dome flats or try morning twilight twilights if overly cloudy
- c) Point telescope towards zenith
 - i) Manually in the dome use the hand paddle
 - ii) Point slightly East to shield from the sun
- d) Change filter **|ACE|**
 - i) {Auxillary Instruments} <Filter Wheel> (B Kron)
 - (1) Use B first while still twilight because least sensitive
- e) Enter name to be recorded in header/Object
 - i) |ACE|<RA>[Name]: twilight flat
- f) Take 1st exposure **|OWL|**
 - i) 5 min after sunset
 - ii) First good exposure is typically

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- (1) 10 min after sunset
- (2) Sun altitude = -4 degrees
- iii) <Exposure Options> **[Open Shutter→check]**
- iv) [Exposure Time]: 4 (always 4 or higher because of shutter issues)
- v) (EXPOSE)
- vi) Check background counts in ds9
 - (1) If counts > 60K too early for sky flats, looking for counts 30-40K
- vii) Keep testing, until counts <40K
- g) Take flats when counts <40K
 - i) Acceptable flats will have counts between 10k – 40k
 - ii) For 4 filters it is possible to get 5 flats for each filter. If less filters are used for science then get as many flats as possible.
 - iii) Adjust exposure times between flats to keep counts up
 - iv) Offset between flats to remove stars **VERY IMPORTANT**
 - (1) |ACE| <Offset> [RA]: 50 (goto) (move)
 - (2) OK to do offset once readout begins
 - v) Shoot 5 acceptable flats in each filter
 - (1) Sample flat sequence (will depend on science)
 - (a) B-Kron
 - (i) Same time for V
 - (b) V-Kron
 - (i) Reduce time 20% for R-Old
 - (c) R-Old (**CAUTION!!!!NOT R-John**)
 - (i) Increase time 20% for I-Kron
 - (d) I-Kron
 - vi) Stop flats when times >180 seconds
 - vii) Sample twilight flat profile
 - (1) B-Kron 4sec to 8sec
 - (2) V-Kron 8sec to 20sec
 - (a) Start with same exposure time as ended in B-K
 - (3) R-Old 12sec to 40sec
 - (a) Shorten exposure time 20% from ending in V-K
 - (4) I-Kron 40sec to 180sec
 - (a) Start with same exposure time as ended in R-O

9) CHECK POINTING (optional)

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- a) Select star from goto catalogue method **|ACE|**
 - i) Rt click In Big Black Box<Open Existing Catalog>(ACE_BSC5.cat) (OPEN)
 - (1) Sort in RA by double clicking RA column
 - (2) Find star near zenith: RA near Sidereal time and DEC close to Latitude MLO +32 deg
 - (3) Click on selected star (click in green area)
 - (4) (GOTO) (MOVE)
 - (5) If catalog won't load try changing directory and find another bright star catalog
- b) Manually enter bright star method **|ACE|**
 - i) Find bright pointing star near to the zenith in Astronomical Almanac
 - (1) 0 to 1 magnitude to assure identification
 - (2) Approx ± 10 degrees off a DEC of 32.
 - (3) <RA> [RA]:xx xx xx.xx; [DEC]:xx xx xx.xx; [EPOCH]:2009.5
 - (4) Be sure to **use appropriate epoch** of the catalogue used for star coordinates
 - (5) (GOTO) (MOVE)
- c) Take exposure **|OWL|**
 - i) [Exposure Time]: 4 --any filter is ok
 - ii) (EXPOSE)
- d) Evaluate targeting position
 - i) Hopefully the star will be in the center--> OK proceed to focus
 - ii) If star is not in center
 - (1) If reasonably placed
 - (a) Manually offset
 - (i) zoom 180
 - (ii) If object is displaced towards left (E) then RA offset is POSITIVE
 - (iii) If object is displaced up (N) then DEC offset is POSITIVE
 - (iv) To center object at edge offset is typically 400
 - (v) If entering negative number enter negative sign before number
 - (b) Automatically Center CCD **|ACE|**
 - (i) {Center CCD) [x]:xxx [y]:yyy (Center Star)
 - 1. x & y are the ccd positions for what you would like to have in the center of the ccd. Find x & y from ds9 screen values
 - (ii) <Offset>(Go to)(Move Offset)

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- (2) If not reasonably placed
 - (a) Hunt around and find it if possible
 - (b) Center CCD as above
 - (c) Reset encoders **|ACE|**
 - (i) "E" (Reset Encoders)
 - (ii) Check that the RA & DEC in the <Positions> box actually change

10) FOCUS

- a) Nominal focus number in **|ACE|{Focus}**
 - i) SUMMER
 - (1) Twilight focus ~ _____
 - (2) Midnight focus ~ _____
 - ii) WINTER
 - (1) Twilight focus ~ _____
 - (2) Midnight focus ~ _____
- b) Find a good star field with lots of 9th mag fields
 - i) Galactic field
 - ii) Globular clusters
 - iii) Open clusters
- c) Expose 10 sec
- d) Evaluate focus **|OWL|**
 - i) IRAF
 - (1) ecl> imexamin axx 1
 - ii) Ds9
 - (1) Over good star type "a" or "r"
 - (2) Identify focus stars to repeatedly use for focus test by placing green circles (click & drag in ds9)
 - (3) Record FWHM- reasonable is 6 pixels at 0.406"/px-> ~2.4"
 - (4) Press ",", to show FWHM, "r" for radial plots, "e" to view contour, "s" to view surface plot (may need to do "a" first)
 - (5) Great FWHM is 4, poor is 8
- e) Adjust focus
 - i) **|ACE| {Focus} [JOG]: xxx (JOG+) or (JOG-)**
 - (1) Adjust first by 100 and not less than 25

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- (2) CAUTION numbers next to GO TO FOCUS are for manually setting. The number to watch is in the header next to FOCUS= or in the left hand column
- ii) Expose again
 - iii) Check focus (FWHM) again with imexamine
 - iv) Repeat and adjust until you find best focus
 - v) Record focus numbers on log sheet
 - f) Advanced focusing technique
 - i) **TURN OFF GUIDER**
 - ii) Turn (open shutter) off on |OWL|
 - iii) Find a sparse star field
 - iv) Set focus in |ACE|{Focus} ~ 100 counts above expected focus
 - (1) First move focus 100 counts below starting focus value then back to starting focus value
 - (2) Record starting focus
 - v) Set offset in |ACE|<Offset>[RA]: 25 [DEC]: 0
 - vi) Set jog focus value 25
 - (1) |ACE|{Focus}[Jog]: -25
 - (a) Record jog focus increment
 - vii) Close shutter in |OWL|{OWL}<Quick Actions> "Iris icon" that is make icon look open
 - viii) Take 240 sec exposure
 - (1) |OWL|{OWL}[Exp. Time]: 240
 - ix) Immediately Open shutter by clicking on Quick Action.
 - (1) |OWL|{OWL}<Quick Actions> "Iris icon" that is make icon look closed
 - x) Count to 10
 - xi) Close shutter with Quick Action as above
 - xii) Count to 6
 - xiii) Offset in RA
 - (1) |ACE|<Offset>(GOTO)(Move Offset)
 - xiv) Jog focus up (25 value set)
 - (1) |ACE|{Focus}(Jog-)
 - xv) When move and focus are complete Open shutter
 - (1) Look at status line under GOTO on ACE
 - xvi) Count to 10

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- xvii) Repeat ix to xiv until time is up
- xviii) Review image in ds9 with imexam and measure the FWHM of each stellar image for one set. Plot FWHM vs Focus value. Interpolate best focus value xxxxx. Increment is from right to left with a + RA offset with ds9 zoom set at 0 degrees.
- xix) Offset focus to compensate for backlash
 - (1) |ACE|{Focus}[Go To Focus]: Starting focus value_(Go to Focus)
- xx) Move to desired focus value
 - (1) Best to make multiple increment moves back to best focus similar steps as during testing
- xxi) Record focus

11) OBSERVE OBJECTS

- a) Find object
 - i) Enter coordinates
 - (1) Manually
 - (a) |ACE|<RA> [RA]: xx xx xx.xx; [DEC]: xx xx xx.x; [EPOCH]: xxxx
 - (b) Check epoch to assure correct
 - (c) Enter object name
 - (i) |ACE|<RA>[Name]: some name
 - (ii) This "some name" will be automatically entered in the header as OBJECT
 - (2) Automatically
 - (a) Use target list in |ACE|
 - (i) Large black box, rt click, choose existing list
 - (ii) Double click on green box next to object name
 - ii) Move telescope
 - (1) |ACE| (GOTO)
 - iii) Take short exposure (~30sec or ? depending on brightness)
 - (1) in R-old band
 - iv) Adjust pointing as necessary
 - (1) Use offsets determined by CHECK ALIGNMENT
 - v) Check placement of bad row (1024-1026)
 - (1) Move telescope if necessary
 - (a) |ACE|{Center CCD}[X]:xxx; [Y]: yyy (center ccd)
 - (i) Enter x & y desired position from ds9

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b) Obtain a guide star

i) |OWL|- dbl click "Max Im DL 5" icon

(1) |OWL|{MaxIm DL} <View> (Camera Control Window)

(a) {Camera Control}

(i) <Setup> (Connect)

(ii) <Setup><Coolers> (On)

(iii) <Expose> [Seconds]: 2 sec

(iv) <Expose>[Continuous→check]

(v) <Expose> (Start)

(2) If bright star in image-> Great, proceed to focus guide star

(a) The guider definitely works better with very bright stars.

(3) If no bright star

(a) |ACE|{Auxiliary...} <XY Stage> Move pickoff camera to find bright star

(i) Press N/S/E/W 50 to 500

1. W-> star to right

2. N-> star down

(ii) Sometimes it is necessary to initialize pick off mirror

1. Particularly if you have to move the pick off mirror a large number of times

2. First record current position numbers

(iii) Always first look for guide star along the top of the X-Y Stage

1. Don't go very close to box, stay above 920 on Y

(iv) Look for a bright star on |OWL|{MaxIM DL} guide star screen

c) Focus guide star

i) |OWL|{Camera Control} <Expose> [Seconds]: 3

(1) <Expose>[Continuous→check]

(2) (EXPOSE)

ii) |ACE|{XY Stage} adjust focus.

(1) 2,000 is significant change. Adjust for best image in guide camera

d) Start guiding

i) |OWL|{Camera Control}

(1) <Expose>(stop)

(2) <Guide> [Seconds]: change as needed

(a) Typically 3-6 sec

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- (3) <Guide> (Expose) (Start)
 - (4) Verify choice of guide star by double clicking in large display.
 - (a) There should be a bulls eye target for the cursor that turns yellow when clicking on the selected guide star. If the cursor isn't visible set the Information Window to be viewable
 - (i) {MaximDL}<VIEW><Information Window>
 - (5) <Guide> (Track) (Start)
 - (6) Verify the star looks reasonable in the small box
 - (7) Nominal settings for guider
 - (a) |OWL|{Camera control}
 - (i) Aggressiveness: X=0.7, y=0.9
 - 1. Maybe necessary to select fine aggressiveness
 - a. <Guide><options>(Fine aggressiveness)
 - (ii) Seconds: 1 <Guide>[Seconds=1]
 - 1. To change exposure time it is necessary to stop and start tracking.
 - (iii) Setpoint -1 <setup><camera1><cooler>[setpoint: -1]
 - (b) |ACE|{Auxiliary}
 - (i) Gain x=50, y=20 <auto guider>
 - (8) Start tracking graph
 - (a) <Guide>(graph)
 - ii) |ACE|{Auxiliary...} <XY Stage> <AutoGuider>(ON)
 - (1) Check that numbers are changing in 2 blue boxes in <Autoguider> tab
 - (2) Best setting for Autoguider Gain RA: 100, DEC: 10
 - iii) It is ok to focus guide star while imaging
- e) Take short science image
- i) First exposure short times (say 30 sec)
 - ii) |OWL|(Expose)
 - iii) Analyze image
 - (1) Load image in ds9
 - (a) Cl> imexam xxxx.fit 1
 - (2) Check image with imexam
 - (a) Place cursor over object of interest & type following imexam command
 - (i) Peak-"z" or "a"

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1. Note "a" removes bias counts automatically
- (ii) FWHM- "a"
 1. Use direct number
 2. Other options "r", ",", "
- (iii) Shape- "e"
- (iv) Surface- "s"
 - (b) Look at peaks, shape and FWHM
 - (3) Check peak counts and use number to adjust exposure times
 - (4) Look at shape and FWHM to assess tracking and focus
- iv) Record info on observing log
 - (1) FWHM of science object, peak value (note z or a such as 6400a), any change to focus, LST
- f) Adjust focus and tracking if necessary
 - i) Focus tends to drop the first couple of hours in the evening (summer and fall any way)
 - ii) Record new focus numbers in observing log
- g) Take full time science image
 - i) Monitor guider performance during imaging by checking graph
 - (1) If saw tooth image on x increase x-aggressiveness
 - (2) If cyclic overshoot (ringing) reduce x-aggressiveness
 - (3) Typical x-aggressiveness range 0.5-0.8
- h) Analyze image as above
- i) Stop guiding
 - i) Before moving to next location
 - ii) |OWL|{Camera Control} <Guide> (Stop)

12) MORNING FLATS

- a) Sky or dome
 - i) Dome flats
 - (1) |ACE|<Telescope> (Close mirror cover)
 - (2) |ACE|<Dome> (Close shutter)
 - (3) |ACE|<Telescope>(Dome Flat Park)
 - (4) |ACE|<Dome>(Azimuth)(Turn off auto dome)
 - (a) Should be azimuth 0
 - (5) Go out onto observatory floor

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- (a) Check telescope is pointing at dome flat mounted on inner dome wall
 - (b) Locate projector on East wall above cabinet
 - (c) Remove fancy dust cover
 - (d) Turn on projector, switch is on South wall
 - (i) Check that projector lamp is set to low
 - (ii) Best to let projector warm up for 10 minutes or more
 - (e) Install filter
 - (i) Filter names are written on top edge of filter
 - (ii) Use carousel remote to advance or rewind to load appropriate filter
 - (f) Neutral Density (ND) cover is in box and installs directly over projector lens
 - (g) Assure flat screen is evenly illuminated
 - (i) If not move projector
 - (6) |ACE|<Telescope> (Open mirror cover)
 - (7) Enter dome flat in name
 - (a) |ACE|<RA>[Name]: Dome Flat
 - (8) Take test flat for each filter to adjust times
 - (a) Look at flat in imexam and zoom to fit
 - (i) Adjust projector if necessary
 - (9) Sample flat sequence (will depend on science)
 - (a) I KG1 20 sec
 - (b) R KG1 10 sec
 - (c) V W80A 8 sec
 - (d) B " 10 sec
 - (e) Adjust Exp. Times to produce counts between 10k and 40k
 - (10) Take (7) multiple flats each filter
 - (a) |OWL|{OWL}
 - (i) <Exposure Options>
 - 1. [Multiple Exposures→check]
 - 2. [Multiple Exposures]: 6
 - 3. [Open Shutter→check]
 - (ii) (EXPOSE)
- ii) Sky flats

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- (1) Move close to zenith (RA~LST, Dec=32)
- (2) Remember to offset between exposures for same filters (50 in RA)
- (3) Start when I-Kron can produce reasonable counts at 180 sec.
- (4) Typically 30 minutes before sunrise
- (5) Shoot for 5 flats in each filter. More are possible if less than 4 filters are used for science. Get as many flats as time permits.
- (6) Take flats
 - (a) Start exposures at 180sec
 - (b) Shorten times as sky becomes brighter
 - (c) Offset between exposures similar to twilight flats
 - (d) Sample sequence (will depend on science)
 - (i) I-Kron
 - (ii) R-Old (**IMPORTANT**- not R-Kron)
 1. Start with same exposure time as ended on I-K
 - (iii) V-Kron
 1. Start with 25% longer time than ended in R-O
 - (iv) B-Kron
 1. Start with same exposure time as ended in V-K
- (7) Stop when B exposure of 4 sec saturates >40k

13) SAVE DATA

- a) On laptop
 - i) {XTERM}
 - (1) Create directory to store
 - (2) Move to directory
 - (3) \$ sftp observer@205.154.251.22
 - (a) (pwd:)
- b) To MOLASUS
 - i) Create directory on MOLASUS
 - (1) \$sftp doug@corelli.sdsu.edu
 - (2) Password:_____
 - (3) sftp> cd MOLASUS/data
 - (4) sftp> mkdir 20130503 (example, use UT date)
 - (5) Exit
 - ii) On OWL
 - (1) Move to image directory

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(2) cd /E/images/dleonard/20130503 (example)

(3) %rsync -tv *fit

doug@corelli.sdsu.edu:/Volumes/Disk35/doug/MOLASUS/data/20130503

c) To macbookpro

i) Create directory on MBP

(a) Cd ~MOLASUS/DATA

(b) Mkdir 20150706

d) On OWL

i) Move to image directory

ii) %rsync -tv *fit

chuckh@172.23.50.223:~chuckh/MOLASUS/DATA/20150706

iii)

14) CLOSE DOWN

a) Clean up workstations

i) |ACE|

(1) <Telescope>

(a) (Close Mirror Covers){Close mirror cover}(yes)

(b) (Track enable){Turn tracking off}(yes)

(2) <Dome>

(a) (Autodome Off){Turn dome tracking off}(yes)

(b) (Close Shutters){Close dome shutters}(close)

(c) Wait for shutters to close all the way

(d) (Azimuth)[Move dome to azimuth]:180

(3) <Telescope>

(a) (Stow telescope) (Stow) (Move telescope)

(i) If it doesn't move. Click (STOP) then use hand paddles to get it near the zenith and try again.

(4) Logoff

(a) <USER>(logoff)

ii) |OWL|

(1) {Controller} "SiDiode temperature" [Set array temp...]: -145 (Run)
(Close)

(2) {Controller} "Parallel Readout" (L) (Run)(Close)

(3) Close all windows except 2 OWL windows

(a) Shut down MiximDL

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(i) Disconnect

1. {Camera Control}<setup>(disconnect)

a. Do not save CCD

(4) <START> (Logoff observer)

(5) Leave monitor on

b) Make notes in observing log book

i) Note general comment on the night, how was seeing, any problems

c) Organize data room and bathroom

d) Remove trash and dispose of in dumpster

e) Lock catwalk door

f) Fill LN2 if necessary and complete LN2 log

g) Unplug telescope fans

h) Unplug guider camera power supply

i) Check pointing of telescope and dome

i) Telescope not to zenith

ii) Dome to 0 or 180

j) Turn off lights downstairs (leave light switch on upstairs)

k) Turn off 4 switches to left of door

l) Exit door

m) Lock door

15) GO HOME

SAMPLE PICTURES

CCD2005 GUIDE



Entry gate



Note placement of chain



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40" Dome



4 switches inside entry door



40" with CCD 2005



Wires connected for fans



Guiding camera power supply
Plugged in

LN2 log on west wall

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East wall LN2 stinger in top
Center cabinet



LN2 stinger attached with tube
Open exhaust pointed over
Plywood



Camera being filled note
Plywood placement. Exhaust
Aimed over plywood



Proper placement of hand paddle

CCD2005 GUIDE



South wall projector switch



Projector with dust cover



Dome flat



Carousel with filters

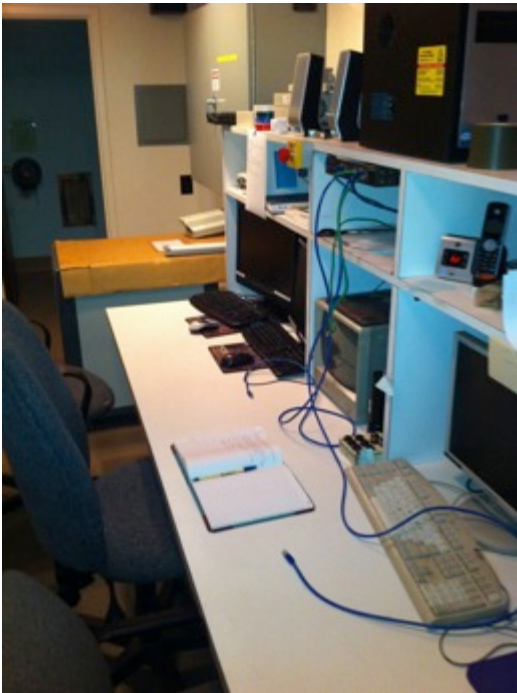
CCD2005 GUIDE



Neutral density filter



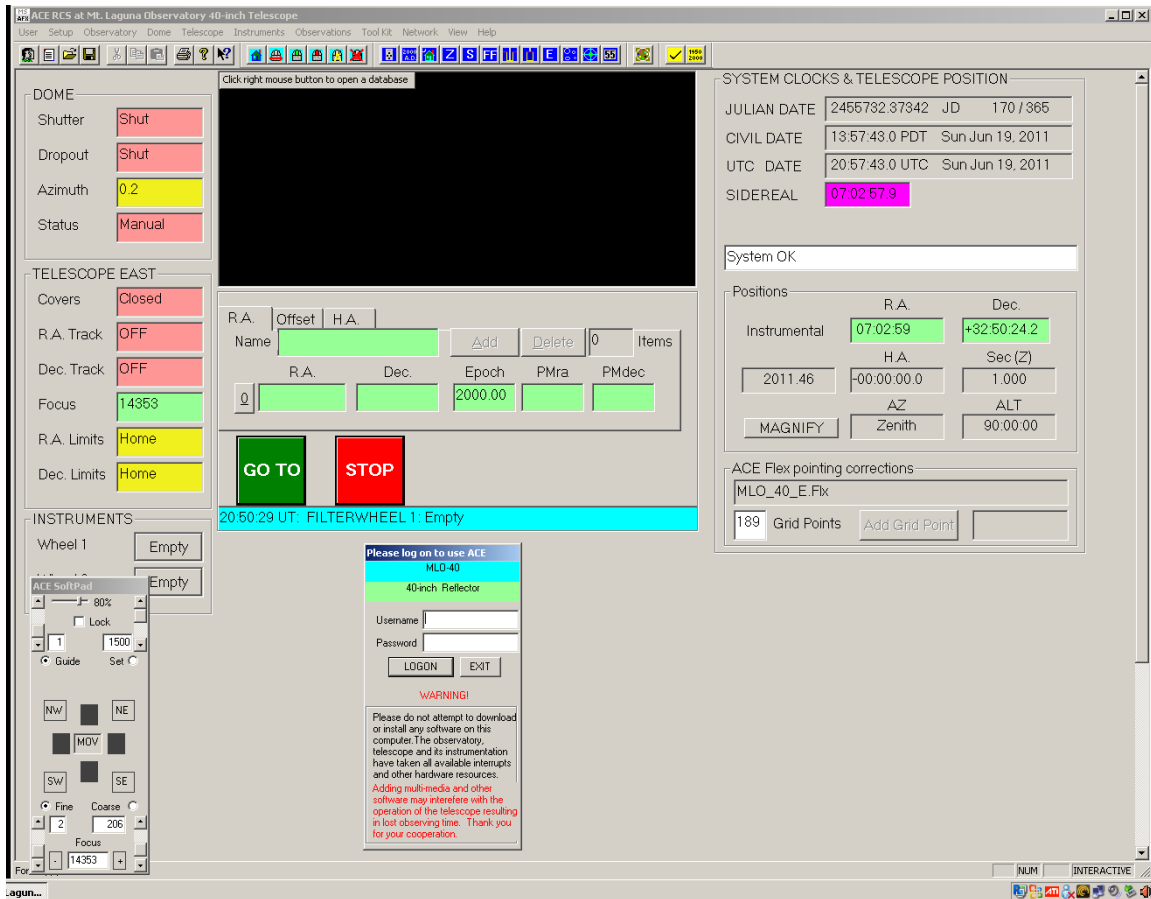
Remote for loading dome flat filters



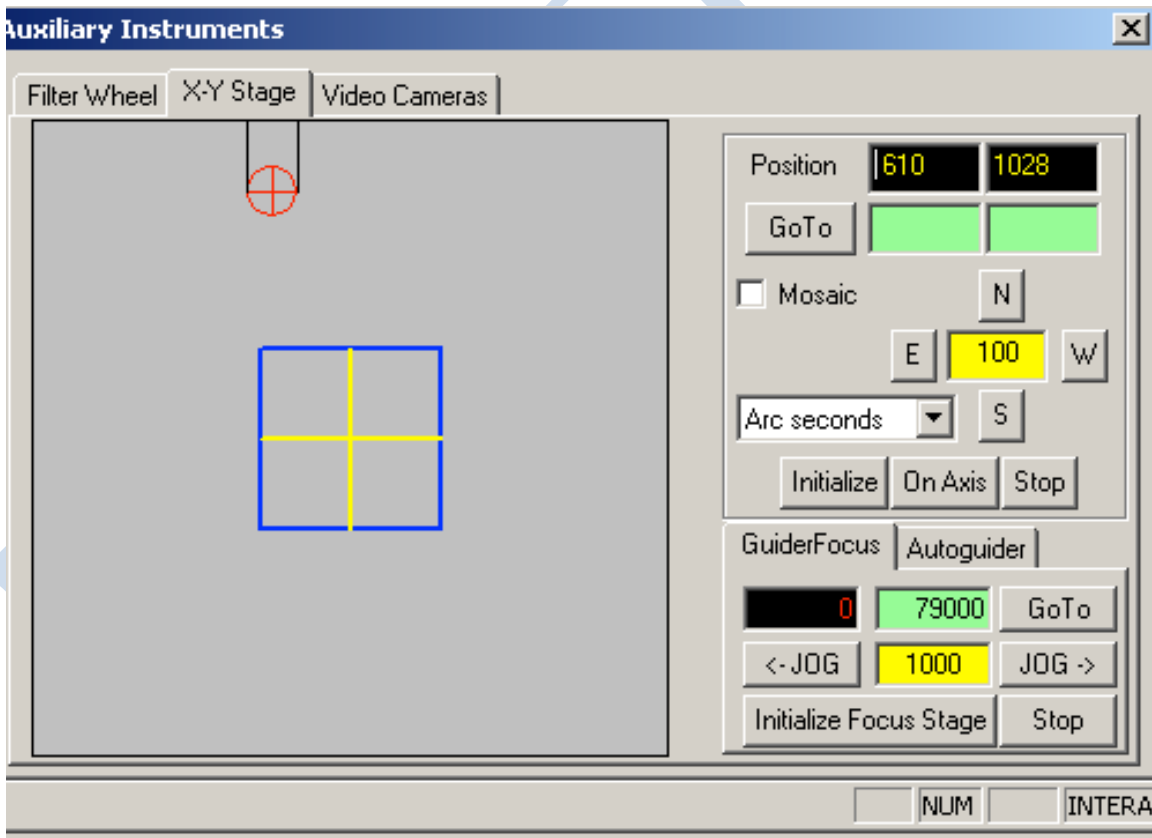
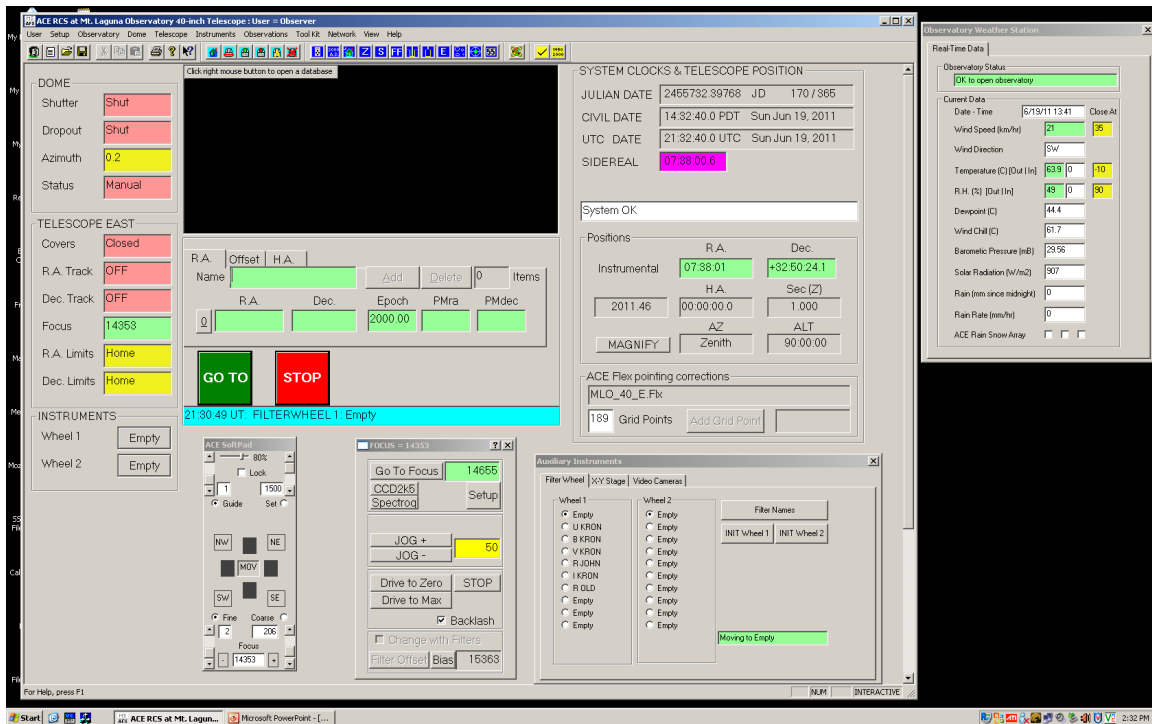
Data room with observing log

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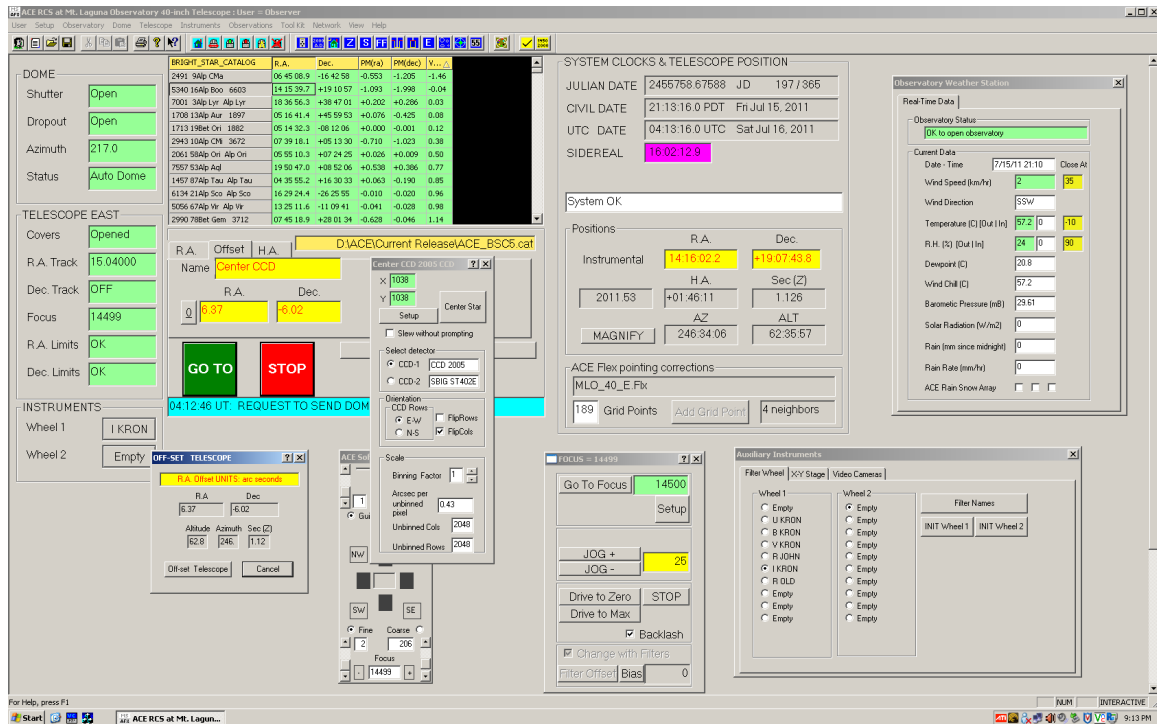
SAMPLE SCREEN SHOTS



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CCD2005 GUIDE

The screenshot displays the CCD2005 software interface. The main window shows a dark image with a green crosshair. The EXPOSE options panel is open, showing settings for Exposure Time (4.0 sec), Read Time (28.43 sec), and various options like Multiple Exposure, Delay Exposure, and Open Shutter. The Array Temperature window is also open, showing a plot of Temperature (C) vs Time (sec) with a red line indicating the temperature trend. The FITS Header window is visible, showing a table of metadata including DATE-OBS, TIMESYS, IRAFNAME, and others.

EXPOSE OPTIONS

Exp Time (s): 4.0 Read Time: 28.43 sec

Exposure Options:

- ☐ Multiple Exposure: 10
- ☐ Delay Exposure (sec):
- ☐ Open Shutter ☒ Beep
- ☐ Synthetic Image ☐ Sub Half

Image File Options:

- ☒ Save ☒ Increment Filename
- Dir: E:\images\deonard\20110716
- File: a31.fit [fit]

Image Analysis:

- ☒ Display

Script: C:\Program Files\Astronomical Res...

Array Temperature

Temperature Value: [-124.8]

Temperature (C) vs Time (sec) plot.

FITS Header

KEY	VALUE	COMMENT
DATE-OBS	2011-07-16T01:08	Date and time of observation
TIMESYS	UTC	Default time system
IRAFNAME	a30	IRAF filename
OBSERVAT	WLO	Observatory
SIDETIME	hh:mm:ss.ss	Local sidereal time
OBJEPOCH	2000	Epoch of object coordinates
AIRMASS	99.99	Airmass
EXPTIME	0.0	Exposure time
HA	hh:mm:ss.ss	Hour angle
TELESCOP	WLO 40"	Telescope
INSTRUME	CCD2005	Instrument
FILTER1	B_Kron	Filter 1
FILTER2		Filter 2
OBJECT	TwilightFlats	Observation title
TELRA		Telescope right ascension
TELDEC		Telescope declination
OBSERVER	Horst	

The screenshot displays the CCD2005 software interface with the Camera Control and Camera Settings windows open. The Camera Control window shows settings for Exposure, Filter Wheel, and Guide Star. The Camera Settings window shows settings for X and Y axes, Auto-Stop, and Manual Calibration. The FITS Header window is also visible, showing a table of metadata.

Camera Control

Exposure: 4.0 sec Read Time: 28.43 sec

Filter Wheel: 10

Guide Star: 10

Auto-Stop: Off

Manual Calibration: Off

Camera Settings

X Axis: Cal. Time Backlash

Y Axis: Cal. Time Backlash

Auto-Stop: Off

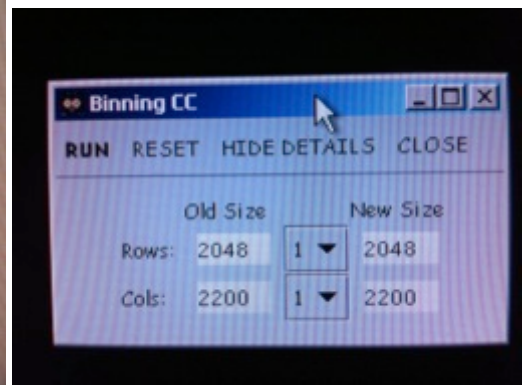
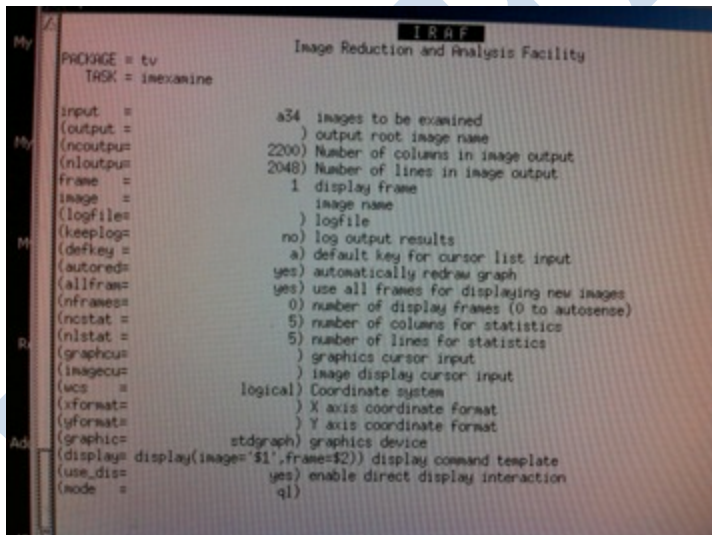
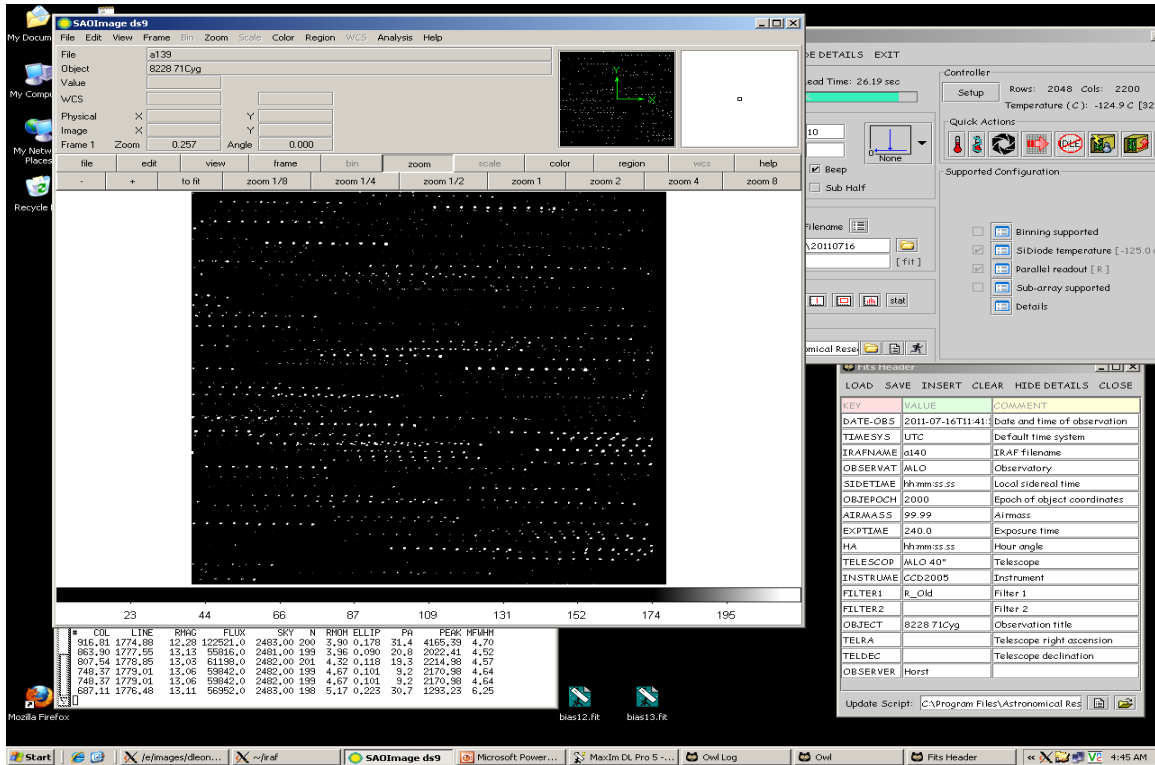
Manual Calibration: Off

Display Mode: Focus

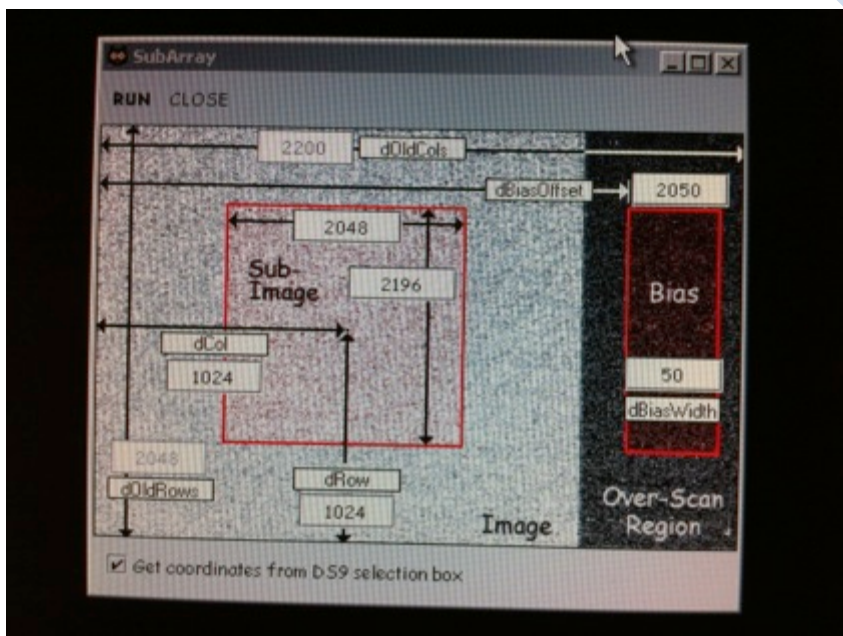
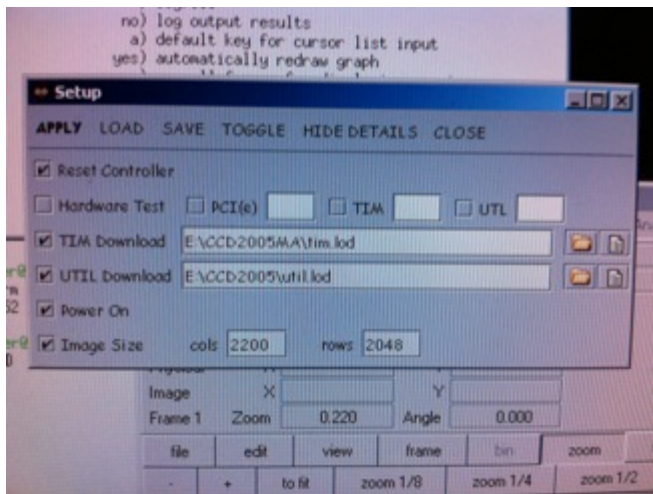
FITS Header

KEY	VALUE	COMMENT
DATE-OBS	2011-07-16T01:08	Date and time of observation
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IRAFNAME	a30	IRAF filename
OBSERVAT	WLO	Observatory
SIDETIME	hh:mm:ss.ss	Local sidereal time
OBJEPOCH	2000	Epoch of object coordinates
AIRMASS	99.99	Airmass
EXPTIME	0.0	Exposure time
HA	hh:mm:ss.ss	Hour angle
TELESCOP	WLO 40"	Telescope
INSTRUME	CCD2005	Instrument
FILTER1	B_Kron	Filter 1
FILTER2		Filter 2
OBJECT	TwilightFlats	Observation title
TELRA		Telescope right ascension
TELDEC		Telescope declination
OBSERVER	Horst	

CCD2005 GUIDE



CCD2005 GUIDE



NOTES

TROUBLESHOOTING

1) DOME STOPS TRACKING

- a) Turn off autodome
 - i) |ACE| <Dome>[turn autodome off→yes]
- b) Rotate dome AMAP each way
- c) Turn on autodome
 - i) |ACE|<Dome>[turn autodome on→yes]

2) DOME NOT WORKING

- a) No dome control at all
 - i) Reset power switch on south wall
 - ii) Wait 20 seconds

3) IRAF

a) On Imexam IRAF keeps returning to ecl prompt

- i) Close IRAF and ds9
- ii) Restart ds9 and IRAF from new xgterm sessions

b) Imexam won't work

- i) Open NOAO package
 - (1) CI> NOAO

4) POINTING THE TELESCOPE

- a) Only adjust pointing if things are way off
- b) Point to a star in the bright star catalog
 - i) Take an image
 - ii) Offset image to center {ACE} <Telescope> (Center CCD)
 - Identify x/y position of object in ds9 to center
 - Input x/y
 - (Center star)
 - <Offset> (Go to)
 - NOTE: it maybe necessary to change the plate scale for the center CCD to work properly
 - iii) Reset encoders
 - (1) |ACE| "E" Icon at top of ACE screen
 - (2) (RESET) watch that RA and Dec change to something "close" to the value set in the [E]
 - (3) Values will only be close due to distortion correction etc.
 - (4) If they don't change try again.

NOTES

- (5) Test settings by pointing to another bright star. Should be nearly centered.
- c) If you can't find a bright star on the chip use old fashioned way
 - i) Use finder scope
 - (a) Climb ladder at telescope
 - (b) Remove lens cap
 - (c) Point to very bright star like Vega
 - (d) Center the star in the finder using hand paddle
 - (a) Note as of 2011.06.08 place the star a bit off to right of center
 - ii) Take an image
 - iii) Follow reset encoders above
 - iv) Replace lens cap on finder
- 5) **DELAY IN EXPOSING**
 - a) Problem- hit expose and OWL waits up to 20 seconds
 - b) Solution- restart OWL. Not necessary to reboot.
- 6) **NOT SEEING FULL IMAGE IN IMEXAM**
 - a) epar imexam
 - i) ncolums 2000
 - ii) nlines 2048
 - b) Restart IRAF
- 7) **AUTOGUIDER**
 - a) Poor focus (Joey Fedrow assisted with this troubleshooting)
 - i) Having major trouble achieving guider focus
 - (1) Initialize guider focus on |OWL|{Aux. Inst}<XY Stage>
<Guider Focus>(Initialize Focus Stage)
 - (2) Set focus value to 79000 (Go To)
 - (3) Use jog to make minor adjustments
- 8) **CAMERA MIRROR DRIVE NOT WORKING**
 - a) Necessary to reset drive motor controller. Beige cord coming out of wall cabinet into server rack. Plugged into APC. From behind, unplug, count to 10, and replug.
- 9) **PERIODIC ERROR NOT WORKING**
 - a) If large sinusoidal on guide camera graph
 - i) Check that PEC is on
 - ii) |ACE|<SETUP> (check PEC is on)

NOTES

iii) If PEC is off restart ACE

10) **AUTOGUIDER NOT WORKING**

1. Check communications
 - a. |OWL| open "MaxIm DL"
 - b. [MaxIm DL]<View> (Camera Control Window)
 - c. {Camera Control} <Guide> (Settings)
 - d. {Guider Settings} <Settings> {Autoguider Output}
 - i. Control Via: Mt. Stromlo
 - ii. COM Port: COM1
 - e. {Guider Settings}<Setup> (OK)
2. Try restarting ACE and OWL programs
 - a. Final attempt is to reboot both computers
3. Reboot sequence
 - a. Shut down both |ACE| and |OWL|
 - b. Restart |ACE|
 - c. Restart |OWL|

NOTES

1) **DOCUMENT LEGEND**

- a) |COMPUTER|
- b) {Window}
- c) <tab> or <section>
- d) [user input field]
- e) user input data
- f) user determined values
- g) (button)
- h) "icon"

2) **DS9 window orientation:**

- a) North is DOWN; East is RIGHT
- b) To make "normal" <Zoom> 180
- c) Moving telescope in +RA moves objects to left in ds9
- d) Moving telescope in +DEC moves them down in ds9
- e) Very large move in either direction is 500

3) **Offset move in ACE**

- a) RA move is in arc seconds. ($\Delta \text{RA} * \cos(\text{DEC})/15$)
- b) Dec move is ΔDec (degree seconds)

NOTES

4) EDITING IMAGE HEADERS

- a) Look at header
 - i) `$imhead foo.fit lo+`
- b) Edit headers single
 - i) `$hedit foo.fit`
- c) Edit list
 - i) Create list in vi of objects or redirect list `%ls > list`
 - ii) `$hedit @list`

5) Editing object list for [ACE]

- a) Stored in [ACE] D/ACE/Current Release
 - i) Use thumb drive to move large file
- b) Edit on ACE with notepad for small edits
 - i) Note: change to view all file types
- c) Tips for creating file
 - i) First row the headings must have no spaces in each title
 - ii) Columns must be left justified with heading
 - iii) Don't use tabs use spaces
- iv) Create large files in spreadsheet
 - (1) Save as windows.csv
 - (2) Open up in text edit and save as Unicode 8
 - (3) Convert to fixed width
 - (e) `Awk -F, `{printf("%-25s %-10s %-10s %-8s\r\n", $1, $2, $3, $4)}` filename.csv.txt > filename.cat`
 - (f) Above command is for: NAME RA DEC V